



Day 18: Amazon S3: Object Storage and Management

[General Interview Questions]

◆ 1. What is Amazon S3 and how does it differ from EBS?

Answer:

Amazon S3 is **object storage**, designed for storing unstructured data, while EBS is **block storage** for EC2. S3 is accessed via API/URL and doesn't attach to EC2.

◆ 2. What happens when you enable versioning in an S3 bucket?

Answer:

Each time an object is uploaded with the same key, **a new version is created** instead of overwriting the previous one.

◆ 3. How does S3 ensure high durability of objects?

Answer:

S3 provides **99.999999999% (11 9's) durability**, by **replicating data across multiple facilities** within the selected region.

◆ 4. What is the purpose of S3 Static Website Hosting?

Answer:

It allows you to **serve static content (HTML/CSS/JS)** via an S3 bucket using **HTTP endpoints** (not HTTPS without CloudFront).

◆ 5. What are the differences between SSE-S3, SSE-KMS, and SSE-C?

Answer:

- **SSE-S3:** Amazon manages the keys
- **SSE-KMS:** Uses KMS for key management and audit
- **SSE-C:** You provide your own keys

◆ 6. How can you allow another AWS account to access your S3 bucket securely?

Answer:

Use a **bucket policy** with a **cross-account principal**, optionally combined with IAM role-based access.

◆ **7. What is the use of Pre-signed URLs in S3?**

Answer:

To provide **temporary, secure access** to private S3 objects without making them public.

◆ **8. What is the benefit of using Multipart Upload in S3?**

Answer:

It allows for **parallel uploading of large files**, **resumable uploads**, and **greater reliability** for large objects (up to 5TB).

◆ **9. What happens if you upload the same key to a versioned bucket?**

Answer:

A **new version ID** is created, and the older version is retained unless deleted manually.

◆ **10. How does S3 Lifecycle configuration help optimize cost?**

Answer:

It **automatically moves objects** to cheaper storage classes (e.g., Glacier) or **deletes** them after a defined period.

◆ **11. How is S3 different from Amazon Glacier or S3 Glacier?**

Answer:

S3 is designed for frequent access, Glacier is for **archival** storage with **longer retrieval times** and **lower cost**.

◆ **12. Can S3 buckets be replicated across regions?**

Answer:

Yes, using **Cross-Region Replication (CRR)**, enabled by **versioning** and a destination bucket in a different region.

◆ **13. What is a delete marker in versioned S3 buckets?**

Answer:

A special object that **marks the latest version as deleted**, but previous versions still exist unless removed.

◆ **14. How does Server Access Logging help in S3?**

Answer:

It provides **detailed access records** (who accessed, what action was taken) for security auditing and monitoring.

◆ **15. What steps are needed to host a static website in S3 with a custom domain?**

Answer:

- Enable **Static Website Hosting**
 - Upload website files
 - Make bucket **public** or use **CloudFront**
 - Map domain in **Route 53** with S3 endpoint or CloudFront
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◆ **1. What are the key differences between S3 Standard, IA, and Intelligent-Tiering?**

Answer:

- **Standard:** Frequent access, low latency
- **IA (Infrequent Access):** Lower cost, retrieval fees
- **Intelligent-Tiering:** Monitors usage and moves data automatically to lower-cost tiers

◆ **2. When should you use S3 Glacier or Glacier Deep Archive?**

Answer:

For **archival data** where retrieval is rare, and cost is the primary concern (e.g., compliance, backups).

◆ **3. What is the requirement to enable replication between buckets?**

Answer:

Versioning must be enabled on both source and destination buckets. A replication role with the required permissions is also needed.

◆ **4. Can S3 replicate encrypted objects?**

Answer:

Yes, but additional configuration is needed if using **SSE-KMS**. You must provide access to the KMS key in the destination account.

◆ **5. What's the difference between Cross-Region and Same-Region Replication?**

Answer:

- **CRR:** Replicates to a bucket in a **different region** (for disaster recovery)

- **SRR**: Replicates within the **same region** (for log aggregation, testing)

◆ 6. What are ACLs in S3? Are they still recommended?

Answer:

ACLs (Access Control Lists) define **object-level permissions** but are considered **legacy**. Bucket policies are now preferred for most use cases.

◆ 7. How do bucket policies differ from IAM policies?

Answer:

- **Bucket Policies**: Resource-based, attached to the bucket
- **IAM Policies**: Identity-based, attached to users, groups, or roles

◆ 8. What happens if both a bucket policy and an IAM policy apply to a user?

Answer:

The effective permission is a combination of both. If **either denies access**, access is **denied**.

◆ 9. What is “Block Public Access” in S3 and why is it important?

Answer:

It enforces restrictions to **prevent accidental public exposure** of buckets/objects regardless of policy or ACL settings.

◆ 10. What does the Metrics tab in S3 help you monitor?

Answer:

It shows **storage usage**, **number of requests**, **error rates**, and data retrieval activity, useful for cost and performance insights.

◆ 11. How can you track object-level metrics in S3?

Answer:

Enable **CloudWatch Request Metrics** and use **Storage Class Analysis** under the Management tab.

◆ 12. What are delete markers in versioned buckets and how do they affect replication?

Answer:

They mark the object as deleted. By default, they are **not replicated** unless explicitly enabled in replication rules.

◆ 13. What happens if you upload a new object with the same name in a versioned bucket?

Answer:

A **new version** is created; the old one is retained and accessible unless deleted manually.

◆ **14. Can you replicate objects between accounts?**

Answer:

Yes. You need to **configure a replication IAM role**, enable versioning, and grant permissions in **both accounts**.

◆ **15. What are common use cases for same-region replication?**

Answer:

- Maintaining copies for **testing** or **analysis**
- Creating **multiple access points**
- **Log aggregation** or **auditing** in centralized buckets